WASTEWISE-ML · RISK REGISTER AND HAZARD ANALYSIS

WasteWise-ml risk register and hazard analysis

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-25	Amish Chadha	First risk register, preliminary hazard analysis and design FMEA (kit 1.8 and 1.9.1 upgrade, stage P3)

WasteWise-ml risk register and hazard analysis

The top risks are a hazardous item graded as sortable (a cell crushed in a bale can start a fire) and a model that never reaches its accuracy targets because the field data are harder or scarcer than assumed. Both are controlled by design (a separate high-recall hazard output, an honest "Unsure", a hazard box) and by evaluation before use (a 2,000-image field set with 400 hazards), but neither can be closed on paper. Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection is scored 1 (almost certain to be caught) to 5 (unlikely to be caught).

> **Safety:** WasteWise-ml guides sorting; it does not make any item safe to handle and does not certify material. The station handles sharps, broken glass, lithium cells, chemical containers and biological waste, and the phone, power bank and shutter button carry lithium cells.

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
R1	No field partner	The first candidate (SWaCH) is not approached or declines	No grade list, no field photos, no evaluation; the project stops at TRL 3	3	5	Project lead	Approach SWaCH through the Helpful Engineering network once TRL 4 is opened; keep a second candidate cooperative
R2	Accuracy or coverage short of target	Field photos harder than assumed; learning curve flatter (M4); stream differs (M5)	PR-001 to PR-004 not met; pickers lose trust	3	4	ML lead	Measure the learning curve on the first 2,000 photos before collecting more; keep "Unsure" a normal answer; publish per-class results
R3	Too few hazard photos	Hazards are about 3 % of the stream; 2,400 needed for training	Hazard recall below 98 % (PR-003)	4	5	ML lead	Photograph hazards on purpose at drop-off points, by trained people; 400-item hazard evaluation (N1)
R4	Consent or ownership dispute	Protocol not agreed before collection	Data withdrawn; harm to the partnership	2	5	Data lead	Written protocol with the partner before any photo (D6, PR-014); DPDP Rules timeline (REG G2)
R5	Dataset license blocks published weights	TACO images under mixed licenses; non-commercial sets	Weights cannot be published under MIT (PR-013)	3	3	Data lead	ZeroWaste excluded (N2); TACO filtered image by image; field data owned with the partner
R6	Phone choice changes the rig	Lens offset or focal length differs from the assumed 26 mm, centred lens	Picture off the mat (PR-019); arm and tray remade	3	2	Design lead	Buy and measure the reference phone before cutting the arm (confirm items 1 and 2)
R7	Labeling pay below a fair rate	Planning figure of USD 6 an hour used as the rate	Exploits the people the project serves	2	4	Project lead	Rate set with the partner at or above the local living wage (D7)
R8	Concept choice reopened	Amish chooses option B	Drawing, BOM, build plan and requirements redone	1	3	Project lead	Decision D1 in WML-DEC-001 with the scoring of WML-OPT-001

Preliminary hazard analysis

Table 2. Hazards at the sorting station with the rig.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Missed hazard graded as sortable	Model error on a cell, sharp or chemical container	Fire in a bale or sack; cut or needle-stick downstream	5	2	Separate hazard output tuned for recall (PR-003); a hazard result can never be overridden into a grade; buyer's check stays final; 400-item hazard evaluation before use	Medium: depends on field recall (R3)
H2	Loose cell shorts in the hazard box	Terminals touch the steel walls, other cells or cans	Heat, thermal runaway, fire	4	2	Terminals taped, cells laid in the 50 mm sand bed clear of the walls, tape roll in the box (F3, PR-018); box lidded, away from paper and film	Low
H3	Cell in thermal runaway in the hazard box	Damaged or crushed cell put in the box	Fire, smoke	4	2	Steel box and sand bed contain one phone-size cell (89 K mean sand rise, WML-CAL-001 section 14); larger packs to the partner's fire guidance; box outdoors or near the door	Medium for packs larger than a phone's
H4	Cuts from sharps and broken glass	Handling items; reaching under the phone	Cuts, infection	3	3	Cut-resistant gloves; the app never asks anyone to hold an item closer; items lie still on the mat; sharps container	Medium (inherent to sorting)
H5	Phone or power bank overheats	Rig in direct sun (about 85 C)	Swollen or burning cell, throttled phone	4	2	Rig stands in shade (BLD safety stop S4, PR-020); power bank shaded, off hot metal; never charge a swollen pack	Low
H6	Repetitive reach to the phone	Screen 1,215 mm high, above the shoulder of short users	Shoulder strain	2	3	Shutter button at the bench edge (F1, PR-015); tap only as fallback	Low
H7	Coin cell swallowed	Spare CR2032 cells for the shutter button within reach of small children at the site	Internal burns	4	1	Spare cells kept out of reach, with the partner's site rules; prefer a button whose cell cover needs a tool	Low

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
						or a twist (to confirm when bought)	
H8	Sharp edges on the made stand	Cut aluminium edges and tray corners	Cuts during building and use	2	3	Deburr and file every edge; tray corners rounded (BLD safety stop S2)	Low
H9	Wrong grade costs the picker money	Misclassification of a sortable item	Lower price or a dispute with the buyer	2	3	"Unsure" as a normal answer; grades are advice; per-class error rates in the model card	Low
H10	Personal data in photos	A face or identifiable person in a field photo	Privacy harm	3	2	Frame is the mat; no faces kept; opt-in upload; protocol with the partner (PR-014, REG G2)	Low

Design FMEA (first pass, P3)

Table 3. Failure modes of option A. RPN = severity x occurrence x detection.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	Hazard output of the model	Flag every hazard	Hazard missed	Fire or injury downstream (H1)	Rare class, too few training photos	5	3	4	60	Hazard photos collected on purpose; 400-item evaluation; threshold set for recall; buyer's check
F2	Class and grade heads	Name the class and grade	Confident wrong grade	Lost value, dispute	Grades not visible in a photo; field shift	3	3	3	27	Per-class thresholds; "Unsure"; model card; retraining loop
F3	Camera picture	Fall on the mat	Picture runs off the mat	Bench shows at the edges; lower accuracy	Lens shorter than 25.76 mm equivalent; phone tilted or turned in the tray; bump loose in the window; lens off centre	2	3	2	12	F5 (32 mm window, phone against the far lip); measure the phone; lens at 420 mm proposed (D2); first check "Picture on the mat"; arm cut to the lens offset
F4	Phone tray	Hold the phone level, bump in the window	Phone slides or tilts	Picture shifts; phone falls on the mat	Phone wider or thicker than the tray; bump larger than the window	3	2	2	12	Confirm item 1; tray made after the phone is bought
F5	Post and arm joints	Keep the camera still	Joint loosens	Picture drifts	Nyloc nuts not tightened; wood screws strip in the plywood	2	2	2	8	Hold point in step 5; nyloc nuts; check at the start of each shift
F6	G-clamps	Hold the board to the bench	Clamp slips	Stand moves; picture shifts	Bench top thicker or thinner than the clamp's range; bench edge not square	3	2	2	12	Confirm item 3; clamps hand tight plus a quarter turn
F7	USB cable	Power the phone	Cable pulled or chafed	Phone stops charging; plug damage	Cable caught by the user; rubbing on a bolt head	2	2	2	8	Cable along one edge of the arm top, clear of the bolts (F4); ties; 1.5 m with slack
F8	Shutter button	Trigger the scan	Button does not trigger	User falls back to tapping the screen	Cell flat; pairing lost; app does not get the key press	2	3	1	6	Confirm item 5; cell life about 560 shifts; screen tap as fallback
F9	Result screen and voice	Tell the user the bin	Result misread	Item in the wrong bin	Grazing view of the flat screen; noise drowns the voice	3	2	3	18	Full-screen colour and 40 mm bin number (F1, PR-016);

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
										voice by default; trials at TRL 4
F10	Sorting mat	Plain, flat background	Mat swells or cups	Uneven background, glare	Water through an unsealed face or edge	2	2	2	8	Sealed all over (F2, PR-017)
F11	Hazard box	Contain hazards	Short or fire in the box	Fire at the station (H2, H3)	Bare terminals; large pack	5	2	3	30	Taped terminals and sand bed (F3); lid on; box away from combustibles; fire guidance for large packs
F12	Phone	Run the model all shift	Overheats or flat battery	Scans stop	Direct sun; screen left on	3	2	2	12	Shade (S4, PR-020); screen sleeps between scans (PR-010); power bank

The highest RPNs (F1, hazards missed, 60; F11, hazard box fire, 30; F2, confident wrong grade, 27) belong to the model and the hazard route, not the made rig. They are tested at TRL 4 with the field evaluation set and the site briefing; nothing in this table is verified yet.